

FOULUM, Denmark

WMO: 060690

Lat: 56.4931N

Lon: 9.5708E

Elev: 58

StdP: 100.63

Time Zone: 1.00 (EUC)

Period: 94-17

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-9.8	-7.4	-11.9	1.4	-9.0	-9.7	1.7	-6.8	13.2	5.1	12.0	5.0	1.9	0	0.634

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	8.0	25.6	18.1	23.7	17.4	21.7	16.6	19.1	23.4	18.1	22.0	17.2	20.8	4.5	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.6	12.7	20.9	16.5	11.8	19.8	15.7	11.2	18.8	54.6	23.4	51.2	22.1	48.5	20.9	22.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
10.5	9.3	8.2	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	8.0	0.6	0.9	3.3	6.9	10.7	13.5	15.9	16.2	13.0	9.2	4.3	1.2
	DBStd	6.52	4.18	3.79	3.31	3.45	2.85	2.70	2.68	2.40	2.40	2.92	3.45	4.31
	HDD10.0	1384	293	253	209	105	26	2	0	0	3	48	172	272
	HDD18.3	3787	551	487	467	342	238	147	85	75	161	282	421	531
	CDD10.0	664	0	0	1	13	47	108	184	193	93	25	1	0
	CDD18.3	23	0	0	0	0	0	3	11	9	0	0	0	0
	CDH23.3	187	0	0	0	1	3	26	95	62	1	0	0	0
	CDH26.7	22	0	0	0	0	0	3	12	6	0	0	0	0
Wind	WSAvg	3.9	4.5	4.7	4.4	3.8	3.8	3.5	3.2	3.3	3.5	3.8	4.0	4.0
Precipitation	PrecAvg	753	67	49	51	41	50	64	65	76	70	82	69	69
	PrecMax	938	135	121	108	93	89	123	101	141	158	174	146	138
	PrecMin	503	7	13	8	5	9	6	20	25	24	17	24	20
	PrecStd	98	33	25	26	23	20	26	27	31	35	34	29	35
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.6	9.1	13.9	21.4	23.0	26.8	28.4	27.5	22.7	16.7	12.8	11.0
		MCWB	8.6	7.7	9.1	14.3	15.5	17.9	19.3	18.6	16.9	14.3	11.9	10.2
	2%	DB	8.0	7.8	11.1	17.8	20.1	23.2	25.9	25.2	20.3	15.5	11.0	9.8
		MCWB	7.2	6.7	8.2	12.3	13.9	17.6	18.5	17.8	16.1	13.7	10.0	9.1
	5%	DB	7.0	6.9	9.2	15.2	18.2	20.8	23.9	23.2	18.4	14.4	9.9	8.4
		MCWB	6.1	5.9	7.1	10.7	13.2	16.0	17.8	17.1	15.1	13.0	9.1	7.7
	10%	DB	6.0	6.0	7.7	12.7	16.2	18.8	21.7	21.2	17.0	13.4	8.9	7.0
		MCWB	5.2	5.1	6.2	9.4	12.3	14.8	16.9	16.6	14.4	12.3	8.2	6.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.8	8.0	10.2	15.2	17.0	19.6	21.1	19.9	17.7	15.3	11.7	10.2
		MCDB	9.5	8.8	12.5	19.6	20.6	23.4	25.9	24.6	21.0	16.2	12.6	10.6
	2%	WB	7.4	7.0	8.5	12.6	15.1	17.8	19.6	18.7	16.7	14.1	10.2	9.1
		MCDB	8.0	7.7	10.6	17.0	18.4	21.8	23.9	22.9	19.6	14.9	10.8	9.7
	5%	WB	6.3	6.1	7.3	11.2	13.7	16.6	18.3	17.8	15.7	13.2	9.2	7.8
		MCDB	6.8	6.8	8.8	14.7	17.1	20.5	22.1	21.7	17.9	14.1	9.7	8.4
	10%	WB	5.4	5.2	6.4	9.9	12.5	15.3	17.3	17.0	14.8	12.4	8.2	6.3
		MCDB	6.0	5.9	7.6	12.3	15.7	18.3	21.1	20.2	16.4	13.4	8.8	6.9
Mean Daily Temperature Range	5% DB	MDBR	4.0	4.2	5.7	7.5	8.1	7.8	8.5	8.0	6.6	5.4	4.0	4.0
		MCDBR	4.1	4.4	8.2	11.2	11.7	11.2	12.2	11.1	8.8	5.9	4.3	4.2
		MCWBR	4.1	3.8	5.6	6.6	6.9	6.5	6.1	5.1	5.4	4.6	4.1	4.2
	5% WB	MCDBR	4.0	4.3	7.5	10.6	10.1	10.1	10.4	9.2	7.8	5.4	4.2	4.3
		MCWBR	4.1	4.0	5.5	6.5	6.7	6.3	5.9	5.1	5.1	4.5	4.1	4.3
Clear-Sky Solar Irradiance	taub	0.310	0.332	0.352	0.388	0.381	0.373	0.391	0.392	0.368	0.347	0.325	0.296	
	taud	2.222	2.277	2.303	2.259	2.323	2.365	2.336	2.347	2.401	2.390	2.292	2.209	
	Ebn at Noon	565	716	802	825	857	868	842	814	782	695	547	479	
	Edh at Noon	55	77	96	117	116	113	114	106	87	69	51	43	
All-Sky Solar Radiation	RadAvg	0.45	1.03	2.39	3.98	5.23	5.65	5.16	4.08	2.81	1.43	0.56	0.32	
	RadStd	0.08	0.15	0.22	0.48	0.49	0.47	0.58	0.34	0.29	0.17	0.09	0.04	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (44 neighbors)	+0.37	N/A	N/A	N/A	N/A	N/A	-119	-161	N/A	N/A

Nomenclature: See separate page